Pelican Wireless Systems

Climate Control API Interface Specification

Overview

Pelican Wireless Systems provides an open standards based real-time interface to all of the management features of their Internet Programmable Thermostats and Power Control Modules. The interface uses simple Http GET and POST commands to request settings or change active parameters. All responses are provided using standard XML notation. Transactions are only allowed over Secure Socket Layer (SSL) connections using password protection. These standard protocols are easily implemented by developers on all types of computing platforms using modern scripting and programming languages.

Types of Information

API access is divided into 3 categories: Thermostats, Power Control Modules, and MySites.

1. Thermostat access includes real-time access to all thermostat configurations, active settings, and current readings. In addition, up to 2 years of historical data can be retrieved for reporting and analysis.
2. Power Control access includes real-time access to configurations and active state.
3. MySites access provides a convenient way to obtain summary information. Users can query information about all of the sites they can access.

Provided Software

Because the API uses standard protocols, there is no restriction on what type of systems or software may access the Pelican system. The client software simply establishes a secure (SSL) connection to their private web server address and initiates an http request. To further simplify access to the API, Pelican has developed a Perl module which provides an easy to use wrapper which gives Perl developers an even more simplified interface to the API. This Perl module can also be used by developers as a reference on how to construct valid requests and how to process API responses.

Constructing an API Request

Http requests are constructed with a list of named values. The Pelican API requires 6 elements in each request. These values are:

username – This is the Email ID of a valid Pelican Site Manager user. This Email ID is created using the administrative interface of the Pelican Web App.

password – This is the password for the Email ID specified above. Since the Pelican system only allows SSL connections, the password is automatically encrypted for each request.

request – The value of this parameter should be either “set” or “get”. This identifies the type of request being made. “get” requests are intended to provide the current values of attributes being requested. “set” requests are intended to change the values of attributes.

object – This is the type of object to which the “get” or “set” is being applied. The currently supported object types are:

1. For Thermostats: **Thermostat**, **ThermostatSchedule**, **SharedSchedule** and **ThermostatHistory**.
2. For Power Control Modules: **PowerOutput**.
3. For MySites: **Sites**.

selection – This is a set of attribute/value pairs which should be used as a query match for the “set” or “get” request. In a “get” request, items matching the selection will be returned in the XML reply. In a “set” request, items matching the selection will be modified in the Pelican system. Pairs are separated by semicolons (;) and attributes are separated from their values by colons (:) (See example below). Attribute names are not case sensitive. Attribute values are case sensitive.

value – For “get” requests, this is a semicolon (;) separated list of attributes which are being requested. For a “set” request, this is a semicolon (;) separated list of attribute/value pairs to be modified. The attribute names are separated from their values by a colon (:). Semicolons and colons are invalid characters for either the attribute or the value. Attribute names are not case sensitive. Attribute values are case sensitive.

When using HTTP GET, the 6 required elements must be formatted using standard http notation with the element name as shown above followed by an equal sign and then that elements value. The first element is preceded by a question mark (?) and the elements are separated by the ampersand (&) character. Standard HTTP character escaping is supported. When using HTTP POST, standard encoding is supported.

The web address of the API interface is the full web address of the site followed by “/api.cgi”. Therefore, a valid API request would be as follows:

https://demo.officeclimatecontrol.net/api.cgi?username=myname@gmail.com&password=mypassword&request=get&object=Thermostat&selection=name:TestThermostat;&value=heatSetting;coolSetting;temperature;

Requests for MySites must be directed to: https://mysites.officeclimatecontrol.net/api.cgi

The API Result

The API will return a properly formatted XML version 1.0 reply. The first XML element will be named “response” and will contain a status element named “success” with a numeric result code. If an error is encountered, an element named “message” with a human readable text message will be returned with an explanation of the error. In addition, if the request type is “get” a set of elements will be returned for each item which matched the selection criteria.

Examples

The following XML reply would be returned from the HTTP request shown above:

```
<?xml version="1.0" encoding="UTF-8"?>
<result>
  <Thermostat>
    <name>TestThermostat</name>
    <heatSetting>68</heatSetting>
    <coolSetting>72</coolSetting>
    <temperature>70</temperature>
  </Thermostat>
  <success>1</success>
</result>
```

If “TestThermostat” wasn't found by the system, the following reply would be returned:

```
<?xml version="1.0" encoding="UTF-8"?>
<result>
  <success>0</success>
  <message>No thermostats found matching selection criteria.</message>
</result>
```

User Defined Attributes

There are two types of attributes supported for “Thermostat” objects. The first type are reserved and predefined. These attributes are defined below in the section titled “Object Attributes”. In addition to the predefined attributes, users can define, set and retrieve their own attributes. Any attribute name which is not reserved will be treated like a User Defined Attribute. Defining an new attribute is accomplished by setting its value through a set request. Once set, these attributes can be accessed using get requests. Their is no restriction on the number of user defined attributes or their values. User defined attributes can also be used as part of the selection criteria. For example, if the user wants to manage a group of thermostats through the API, they could define an attribute named “managed” and set its value to “yes” for the thermostats in the user's group. Then they could include “managed:yes;” in their selection criteria to only affect their special group of thermostats.

Thermostat - Object Attributes

Name	**Values**	**Set**	**Description**
name	String	Y	The configured name of the thermostat.
groupName	String	Y	The configured group name of the thermostat.
serialNo	String	N	The thermostat serial number.
system	Off, Auto, Heat, Cool	Y	The active system mode.
heatSetting	Integer	Y	The active Heat Setting.
coolSetting	Integer	Y	The active Cool Setting.
fan	Auto, On	Y	The active Fan Mode.
status	Occupied, Vacant	Y	Normally Occupied. Vacant when vacation schedule is active.
temperature	Decimal	N	The current measured temperature.
humidity	Integer	N	The current measured humidity: % RH. Zero (0) if humidity is not supported.
humidifySetting	Integer	Y	The minimum humidity setting.
dehumidifySetting	Integer	Y	The maximum humidity setting.
co2Setting	Integer	Y	The active CO_2 Setting for Demand Ventilation.
co2Level	Integer	N	The current measured CO_2 level.
setBy	Station, Remote, Schedule	N	Indicates how the active thermostat settings were set.

frontKeypad	On, Off	Y	Thermostat keypad status.
runStatus	Off, Cool-Stage1, Cool-Stage2, Heat-Stage1, Heat-Stage2, Fan, Fan2	N	The currently active Heating or Cooling status.
statusDisplay	String	N	Easily readable indication of what is currently running or “Unreachable” if the thermostat cannot reach the Internet.
auxStatus	On, Off	N	The active Auxiliary Heat status.
slaves	Array	N	List of remote temperature sensors and devices assigned to this thermostat and their values.
description	String	Y	Free form description stored with configuration.
cycleRate	Integer	Y	Configured target cycle rate.
anticipationDegrees	Decimal	Y	Configured anticipation degrees.
calibrationOffset	Decimal	Y	Configured calibration degrees.
heatStages	Integer	Y	Number of heat stages.
coolStages	Integer	Y	Number of cool stages.
fanStages	Integer	Y	Number of fan stages.
HeatNeedsFan	Yes, No	Y	Turn on Fan during heating.
temperatureFormat	Fahrenheit, Celsius	Y	Active temperature format.
systemType	Conventional, Heat Pump	Y	Active system type.
reversingValve	String	Y	One of: Heat or Cool. Energizes reversing valve in this mode. Only valid for systemType of Heat Pump.
minHeatSetting	Integer	Y	Lowest user settable heat setting.
maxHeatSetting	Integer	Y	Highest user settable heat setting.
minCoolSetting	Integer	Y	Lowest user settable cool setting.
maxCoolSetting	Integer	Y	Highest user settable cool setting.
fanCirculationTime	Integer	Y	Minutes/hour to circulate the fan.
heatingType	String	Y	Size of heating system.
heating2Type	String	Y	Size of second stage heating system.
heatingAuxType	String	Y	Size of auxiliary heating system.
coolingType	String	Y	Size of cooling system.
cooling2Type	String	Y	Size of second stage cooling system.
humidityControl	String	Y	One of: None, Humidify, Dehumidify, Full Control, Cool Only.

minSafeHumidity	Integer	Y	Lowest allowed humidity before notification is sent.
maxSafeHumidity	Integer	Y	Highest allowed humidity before notification is sent.
allowKeypadDisable	Yes, No	Y	Display keypad On/Off button.
weight	Integer	Y	Weighted value when used with Remote Sensors
notificationSensitivity	String	Y	One of: Off, Low, Medium, High, Custom. Note: When setting Custom notificationSetpoint and notificationUnreachable should also be set.
notificationSetpoint	Integer	Y	Degrees from setpoint before receiving notification. Forces notificationSensitivity to Custom.
notificationUnreachable	Yes, No	Y	Send notification if thermostat becomes unreachable. Forces notificationSensitivity to Custom.
minSafeTemp	Integer	Y	Lowest allowed temperature before notification is sent.
maxSafeTemp	Integer	Y	Highest allowed temperature before notification is sent.
gateway	String	N	The serial number of the gateway providing Internet access to this thermostat.
version	String	N	The firmware version.
installDate	Date/Time	N	ISO 8601 Formatted Date Time indicating when the thermostat was first installed at the site.
schedule	On, Off, *Name*	Y	Whether the schedule is active or the *Name* of the shared schedule.
scheduleName	String	Y	If using Shared Schedules, the name of the active schedule.
scheduleRepeat	Weekday, Weekly	Y	Schedule repeat.
scheduleSetTimes	2, 3, 4, Variable	Y	Number of set times per day in the schedule.
scheduleMultipleSystem	Yes, No	Y	System mode for each schedule set time.
scheduleMultipleFan	Yes, No	Y	Fan mode for each schedule set time.

ThermostatSchedule - Object Attributes

Name	Values	Set	Description
name	String	N	The configured name of the thermostat.
serialNo	String	N	The thermostat serial number.
dayOfWeek	Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Vacation	Y	The day of the week for this schedule set time.
setTime	Integer	Y	The indexed set time for this schedule entry.
startTime	String	Y	The time of day (24 hr format) for this schedule entry.
system	Auto, Heat, Cool, Off	Y	The system mode.
heatSetting	Integer	Y	The heat setting.
coolSetting	Integer	Y	The cool setting.
fan	Auto, On	Y	The fan setting.
keypad	On, Off	Y	The keypad setting.
delete	None	Y	Delete the specified schedule entry.
deleteAll	None	Y	Delete the schedule for the entire day.

SharedSchedule - Object Attributes

Name	Values	Set	Description
name	String	Y	The name of the shared schedule.
scheduleRepeat	Daily, Weekday, Weekly	Y	The schedule repeat. For “Daily” only the “Sunday” dayOfWeek will be used. For “Weekday” the “Sunday” schedule will be used on weekends and the “Monday” schedule will be used for weekdays. For “Weekly” all days of the week are used.
dayOfWeek	Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Vacation	Y	The day of the week for this schedule set time.
setTime	Integer	Y	The indexed set time for this schedule entry.
startTime	String	Y	The time of day (24 hr format) for this schedule entry.
system	Auto, Heat, Cool, Off	Y	The system mode.
heatSetting	Integer	Y	The heat setting.
coolSetting	Integer	Y	The cool setting.
fan	Auto, On	Y	The fan setting.

keypad	On, Off	Y	The keypad setting.
delete	None	Y	Delete the specified schedule entry.
deleteAll	None	Y	Delete the schedule for the entire day.
scheduleDelete	None	Y	The entire schedule will be deleted.

ThermostatHistory - Object Attributes

Name	**Values**	**Set**	**Description**
name	String	N	The configured name of the thermostat.
groupName	String	N	The configured group name of the thermostat.
serialNo	String	N	The thermostat serial number.
system	Off, Auto, Heat, Cool	N	The active system mode.
heatSetting	Integer	N	The active Heat Setting.
coolSetting	Integer	N	The active Cool Setting.
fan	Auto, On	N	The active Fan Mode.
status	Occupied, Vacant	N	Normally Occupied. Vacant when vacation schedule is active.
setback	On, Off	N	Whether a temporary Demand Response setback is active.
temperature	Decimal	N	The measured temperature.
humidity	Integer	N	The measured humidity: % RH. Zero (0) if humidity is not supported.
humidifySetting	Integer	N	The minimum humidity setting.
dehumidifySetting	Integer	N	The maximum humidity setting.
co2Setting	Integer	Y	The CO_2 Setting for Demand Ventilation.
co2Level	Integer	N	The measured CO_2 level.
setBy	Station, Remote, Schedule	N	Indicates how the active thermostat settings were set.
frontKeypad	On, Off	N	Thermostat keypad status.
runStatus	Off, Cool-Stage1, Cool-Stage2, Heat-Stage1, Heat-Stage2, Fan, Fan2	N	The currently active Heating or Cooling status.
auxStatus	On, Off	N	The active Auxiliary Heat status.
slaves	Array	N	List of remote temperature sensors and devices assigned to this thermostat and their values.
timestamp	Date/Time	N	ISO 8601 Formatted Date Time.

startDateTime	Date/Time	N	ISO 8601 Formatted Date Time. Used for selecting the history date range to retrieve.
endDateTime	Date/Time	N	ISO 8601 Formatted Date Time. Used for selecting the history date range to retrieve. A maximum of 30 days of history can be retrieved in a single request.

PowerOutput - Object Attributes

Name	Values	Set	Description
name	String	Y	The configured name of the output relay.
groupName	String	Y	The configured group name of the output relay.
serialNo	String	N	The Power Control Module serial number.
output	On, Off	Y	The active output status.
setBy	Station, Remote, Schedule	N	Indicates how the active output was set.
schedule	On, Off	Y	This is a Set only attribute and enables/disables the schedule.

MySites - Object Attributes

Name	Values	Set	Description
name	String	N	The site name.
domain	String	N	The full web address of the site.
token	String	N	When a token is requested, it can be used to authenticate with the specific site. When querying the site the token must be placed in the “password” field and the “username” must be blank. Tokens are valid for 24 hours. Tokens are useful when the user has different passwords on multiple sites or if the user has been granted access to a site only through MySites “Shared Users”.
gateways	Integer	N	The number of gateways assigned to the site.
activeGateways	Integer	N	The number of gateways actively communicating with the Internet.
devices	Integer	N	The number of Pelican wireless devices installed at the site.
activeDevices	Integer	N	The number of Pelican wireless devices actively communicating with the Internet.
notifications	Integer	N	The number of active notifications for the site.